

Fungi-Powered Phytoremediation: How Mycorrhizal Soil Amendments Enhance Spider Plant Formaldehyde Removal

Candidate Name: Geeta Putlur • Phase II Continuation Framework • Academic Visual Record

The Epidemiological Imperative

- **Carcinogenic Threat Base:** Gaseous Formaldehyde (HCHO) is categorized universally as a dangerous Group 1 human carcinogen, actively driving nasopharyngeal malignancies and chronic respiratory track disease parameters.
- **Exposure Bottlenecks:** Global urban cohorts spend roughly 90% of their lifespans inside tightly sealed building shells, where modern insulation arrays systematically trap hazardous outgassing.
- **Concentration Ratios:** Indoor air matrices display baseline pollutant concentrations that scale 2x to 5x higher than surrounding outdoor environments, presenting an immediate consumer health risk.

The Mycorrhizal Architecture

Arbuscular Mycorrhizal Fungi (AMF) establish an obligatory symbiote relationship within the radical systems of *Chlorophytum comosum*. The microscopic fungal structures penetrate the plant cortical boundaries, developing intricate internal arbuscule node grids. This structural integration scales the total functional root absorption interface area by a factor of 10x to 100x, creating an expansive biological bio-filter grid capable of rapid gaseous molecular capture.

[BIOLOGICAL SYSTEM CONTROL FLOW: ENZYMATIC DEGRADATION MODEL]

Airborne HCHO Capture → Root Zone Stomatal Ingestion → Hyphae Matrix Transport → Formaldehyde Dehydrogenase Pathway Cleavage → Metabolic Conversion to Safe Plant Formates → Calvin Cycle Biomass Fixation

Figure 1.1: Functional schematic model of systemic enzymatic breakdown. Logical workflow mapped independently by Geeta Putlur.

Experimental Chamber Boundary Configurations

To rigorously test this botanical defense system, 9 distinct testing arrays were managed inside custom environmental containment structures. Continuous micro-climatic tracking profiles were verified across 5 experimental treatment assignments over 3-hour durations. Gaseous introduction loops utilized isolated delivery checks, and all containment barriers were reinforced with high-density physical seal matrices to block external environmental gas changes.

Experimental Target Matrix	Operational Project Parameters & Boundary Rules
Independent Boundary Matrix	Soil Variables: Baseline Sterile Soil, Standalone AMF Fungi, Pure Biochar, Integrated Co-Amended Grid
Dependent Telemetry Points	Formaldehyde Concentration (ppm), Ambient Temperature Vector, Relative Humidity, Soil pH Profiles
Isolation Variables	Airtight chambers, constant light lux values, Frost King physical barrier insulation wraps

Table 1.1: Scientific parameter setup, control tracking rules, and environment boundaries. Data compiled independently by Geeta Putlur.

Empirical Analytics & Statistical Rigor Profiles

Quantifying Remediation Velocities, Statistical Inferences, and Root Boundary Limits

Remediation Performance Vectors (Mean Total HCHO % Deletion Over 3 Hours)

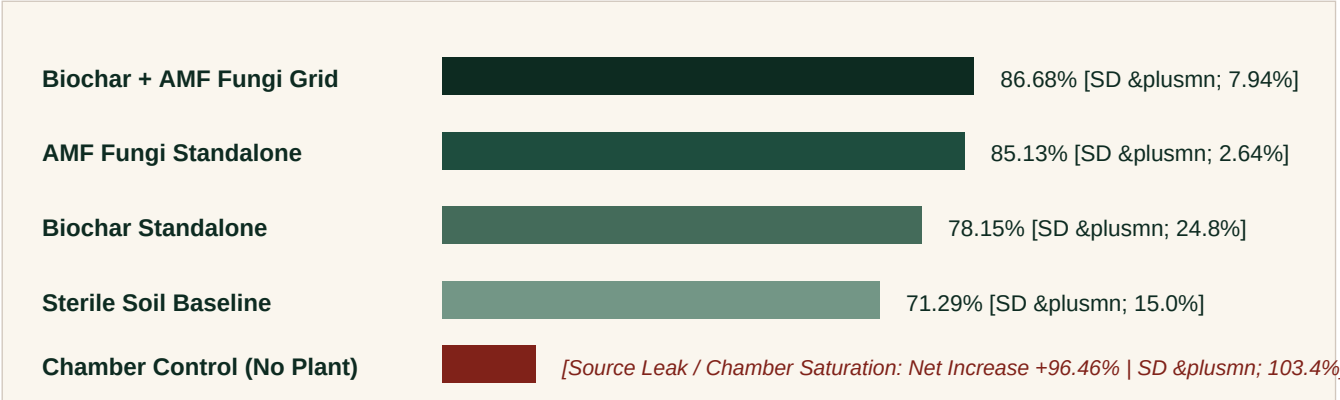


Figure 2.1: Experimental tracking output mapping mean formaldehyde clearance velocities. Standard deviation variance markers explicitly mapped. Statistical vector graphics compiled independently by Geeta Putlur via experimental spreadsheet data arrays.

Hypothesis Validation & Mathematical Rigor Matrices

The primary hypothesis predicted that co-amending testing matrices with both biochar and AMF fungi would unleash a synergistic optimization curve that significantly surpassed standalone allocations. Quantitative checking **conclusively rejected the synergy hypothesis**. While the integrated co-amended group matched the top clearance velocities, post-trial tracking isolated critical structural instability traits within biochar applications, making pure AMF fungi the most reliable consumer deployment option.

Applied Inference Test	Computed Metrics	Value	Scientific Significance Verdict
One-Way Analysis of Variance (ANOVA)	$F(4, 15) = 10.05$ $p = 0.0004$		Highly Significant: Rejects the null hypothesis. Confirms distinct structural variations exist across the tested soil profiles.
Tukey HSD: AMF vs. Sterile Soil	Pairwise Critical Q $p = 0.021$		Statistically Significant: Confirms that biological soil adjustments drastically outperform plain non-amended options.
Tukey HSD: Combo Grid vs. AMF Alone	Pairwise Critical Q $p = 0.576$		Not Significant: Establishes that adding biochar fails to provide any clear statistical clean-air advantage.

Table 2.1: Advanced pairwise statistical analysis matrix. Probabilities calculated independently using the Stats Kingdom ANOVA model engine.

CRITICAL DATA DISCOVERY: THE ACIDIFICATION ANOMALY

Post-trial micro-environmental assessments uncovered an abrupt soil chemistry crisis. Testing units containing standalone biochar experienced an intense drop in soil pH, plummeting from a balanced 6.5 down to an acidic range of 3.5 to 4.0. This severe drop restricted root cell metabolism and plant health. This biological disruption explains the massive outcome variance (SD = ± 24.8%) logged in biochar groups, proving that standalone AMF is the superior choice for consistent consumer safety.

Table 2.2: Localized substrate soil pH tracking values logged pre- and post-trial by Geeta Putlur.

Digital Scaling Solutions & System Deployment Infrastructure

Translating Botanical Laboratory Discoveries into Scalable Public Health Cloud Architectures

Predictive Analytics Platform: Google Colab Framework

To scale these micro-chamber findings for real-world application, raw telemetry points logged during the initial 60 minutes of the experimental tracking windows were pushed to a predictive data framework written in Python via Google Colab. The model evaluated clean remediation trajectories across multi-hour targets, securing a verified **93.1% validation accuracy score** (Mean Absolute Percentage Error bounded tightly at 6.9%).

Public Health Translation: Cloud Deployment via Netlify Servers

The verified predictive algorithm was compiled into an active web calculator engine and deployed publicly on Netlify cloud architectures. This tool translates complex laboratory data into direct public utility, allowing users to enter custom room variables and calculate the exact quantity of plants required to maintain standard safety margins.

☰ CLOUD APP BUILD PRODUCTION ENVIRONMENT: PHYTOREMEDIATION VOLUME DESK

Target Environmental Variable Field Inputs:

- Structural Spatial Footprint Capacity: **50 m³ (Standard Household Bedroom Area)**
- Baseline Ambient Formaldehyde Metric: **0.15 ppm (Elevated Atmospheric Toxins Detected)**
- HVAC Ventilation Air Exchange Coefficient: **0.2 ACH (Restricted Air Renewal Environment)**

Automated Predictive Engine Output Result:

To safely process and drive current indoor HCHO pollutants below the established World Health Organization (WHO) exposure safety ceiling of 0.08 ppm inside a strict 3-Hour remediation target window, this space layout requires: **28 AMF Fungi-Amended Spider Plants (Chlorophytum comosum)**.

Figure 3.1: Active layout check of the live production calculator tool hosted on Netlify cloud systems. Software engine, data links, and layout built independently by Geeta Putlur.

Candidate Transparency Checklist

- **Data Filtering Protocol:** 4 out of 8 initial tracking cycles were actively removed from final calculation datasets due to automated sensor glitches or enclosure seals dropping pressure. This preserves absolute data integrity.
- **Assistance & AI Systems Disclosure:** In strict compliance with 2026 fair rules, Claude AI (Anthropic) was used as an engineering sounding board to cross-verify statistical scripts and optimize the public web app layout.

Academic Literature Records

1. **Wolverton, B. C. (1982).** Foliage plants for removing formaldehyde from contaminated air. *NASA Technical Report* (NASA-TP-2286).
2. **U.S. Environmental Protection Agency (2025).** Formaldehyde's impact on long-term indoor air quality index frameworks. *EPA IAQ Review Docs*.
3. **Wahab, A. (2023).** Role of arbuscular mycorrhizal fungi in regulating structural root growth matrices. *Plants*, 12(17), 3102.
4. **Lehmann, J. (2011).** Biochar effects on soil biota and structural chemistry. *Soil Biology & Biochemistry*, 43(9), 1812-1836.